

Milestone 1 Progress Report

Alethia - AI Symptom-to-Condition Classifier (Group 9)

1. Project Summary

Alethia is a supervised classification system that predicts the most likely medical condition from a binary vector of patient-reported symptoms. For Milestone 1, we built a complete, reproducible data pipeline and trained a K-Nearest Neighbours (KNN) baseline on the Kaggle Disease Symptom Prediction dataset.

The project is strictly an educational demo - it is not a diagnostic tool and does not provide medical advice.

2. What Has Been Accomplished So Far

- Data: Kaggle Disease Symptom Prediction dataset downloaded, plus supporting severity, description, and precaution files staged for later milestones.
- Preprocessing: raw Symptom 1 – Symptom 17 columns converted into binary feature vectors; stratified 80/20 train/test split implemented.
- Baseline model: KNN classifier trained.
- Evaluation: beyond accuracy, we report macro precision, macro recall, and full per-class precision/recall/F1, addressing the TA's "accuracy alone can be misleading" feedback.
- Artifacts saved: trained model, label encoder, and feature column order for reuse in later milestones.
- Safety framing: added a non-diagnostic disclaimer to the README, and documented the dataset-realism limitation the TA flagged.

Result: accuracy = 1.00, macro precision = 1.00, macro recall = 1.00, with perfect per-class scores across all 41 classes. This matches the limitation we already flagged in our design doc: the Kaggle dataset is clean and near-deterministic (each disease maps to a unique, non-overlapping symptom set), so KNN memorizes it trivially. We report this openly rather than treat it as evidence of real-world diagnostic accuracy.

3. What Has Fallen Behind

- Nothing planned for Milestone 1 is incomplete; the KNN baseline and full data pipeline (our Milestone 1 deliverable per the project timeline) are done and reproducible.
- Per TA feedback, we deferred adding a top-3 accuracy metric and Decision Tree overfitting analysis (max depth/pruning vs. Random Forest) to Milestone 2, since Decision Tree, Random Forest, and top-3 output were always scoped for the Model Improvement phase (Jul 2–15).

Challenge: Found the dataset has only ~317 unique symptom patterns duplicated ~15x, causing train/test leakage and inflating our scores to 1.00 — a known issue with synthetic symptom datasets. For Milestone 2, we'll 'deduplicate' for an honest baseline, add severity-weighted features, and build a top-3 output so the model degrades gracefully instead of giving one falsely-confident answer.

4. Any Project Changes

- Evaluation scope (TA feedback): added macro precision/recall and per-class metrics beyond the accuracy/precision/recall originally planned for Milestone 1.
- Framing (TA feedback): reframed the system as an educational symptom-to-condition classifier throughout, with an explicit non-diagnostic disclaimer and a documented dataset-realism limitation.
- No changes to dataset, core AI technique, or overall milestone timeline.